PSU

+15 V switching power supply, noisy.

J1 Input jack — +15V — PWR_FLAG — GND — PWR_FLAG

L2 33u, 0.3 Ohm

LC filter (fc = 1.446 kHz). Choose proper coil resistance for critical dumping.

C15 100u, C21 100u, C35 10u, C16 100u, C62 100u, C64 100u, C66 22u, C68 22u, C70 22u, C55 200n, C63 200n, C65 200n — +15_VDC_IN

U10 ltc3261emse−pbf

+15_VDC_IN — VIN, EN — VOUT — TP17 — C67 6.8n — C12 100n — TP3

+15 V / +12 V LDO, up to 1 A.

LF120_TO220 — VI, VO, GND — U13 — C36 2.2u — TP1 — +12_VDC_A

MODE, RT — NC — L3 33u — C14 10u — U14 — OUT, GND, IN — C39 10u — TP2 — −12_VDC_A

LC filter (fc = 10 kHz).

−15 V / −12 V LDO, up to 1 A.

Inverting charge pump. High ripple (500 kHz, 100 mV p−p). Iout up to 100 mA.

C17 1u

U3 TPS563209DDCR

+12 V to +5 V SMPS, up to 3 A, pretty noisy but for powering external devktis with internal LDOs it really doesn't matter.

+15_VDC_IN — VIN, EN, GND — SW — L1 3.3u — TP6 — +5_VDC_D

C1 10u — C8 100n — VBST — R9 56.2k — C34 22u

FB — R10 10k

U1 REF195GSZ−REEL7

+12_VDC_A — C2 10u, C9 100n — VS, *SLEEP — OUTPUT — TP4 — +5_VDC_REF

TP, NC, NC, TP — C57 1u, C58 100n

High precision LDO, +/− 2 mV maximum accuracy. Iout up to 30 mA.

R18 10k — TP5 — +2.5_VDC_OFF

Zener reference for generating stable Voff = +2.5 V

D10 D_Zener

R52 1k — D9 LED

Simple LED for information purposes.

Input

Phantom power +12 V

EMIF (fc = 7.957 kHz) for maintaining good CMRR (TL07x has 80 dB for 100 kHz). Clamping to +12 V.

INPUT_MIC1 XLR3 — MIC_IN+, MIC_IN− — R2 680, R4 680 — C37 22u, C38 22u — R7 2k, R8 2k — D1 BAT46W, D2 BAT46W — TP7, TP8 — R5 10k, R6 10k — C25 10n, C27 10n

Differential amplifier (gain = 20 dB)

U2A TL074 — R16 100k — R12 100k — TP9

Sixth order butterworth lowpass (fc = 500 Hz)

R1 47k — R3 100k — C3 5.1n — U2B TL074 — C4 4.2n

R11 47k — R13 120k — C10 6.8n — U2C TL074 — C11 2.7n

R19 47k — R22 140k — C19 1.8n — U2D TL074 — C20 1n

MIC_IN_FILT — TP15

Inverting amplifier, unity gain with +2.5 V offset.

C5 100n, C13 100n — U2E TL074 — C6 100n, C22 100n

U4 TL074 — R24 100 — D3 BAT46W, D4 BAT46W — TP13 — MIC_IN

+2.5_VDC_OFF — MIC_IN_FILT — C60 100n — R21 1k, R23 1k — C23 10n

RC (fc = 159.154 kHz) for better stability and cutting possible TL07x noises.

Same design as INPUT_MIC circuitry.

ERROR_MIC1 XLR3 — MIC_ER+, MIC_ER− — R15 680, R17 680 — C31 22u, C32 22u — R25 1k, R26 1k — D5 BAT46W, D6 BAT46W — TP10, TP11 — R27 10k, R28 10k — C33 1n, C40 1n

U5A TL074 — R31 100k — R29 100k — TP12

C44 5.1n — U5B TL074 — C43 4.2n

R33 47k — R35 100k — R36 47k — R37 120k — C46 6.8n — U5C TL074 — C45 2.7n

R38 47k — R39 140k — C59 1.8n — U5D TL074 — C47 1n

MIC_ER_FILT — TP16

C26 100n, C29 100n — U5E TL074 — C28 100n, C30 100n

U9 TL074 — R34 33 — D7 BAT46W, D8 BAT46W — TP14 — MIC_ER

+2.5_VDC_OFF — MIC_ER_FILT — C41 100n — R30 1k, R32 1k — C42 27n

Output

RC and CR combination for cutting possible noises and constant component removal.

Gain = 20 dB

DAC — R53 1k — TP18 — C61 100n — C72 1n, R14 200k

U7A TL074 — R40 2k, R41 18k — R42 47k — R43 100k — C48 4.2n

C49 5.1n — U7B TL074

Same filter as input.

R44 47k — R45 120k — C52 6.8n — U7C TL074 — C51 2.7n

R46 47k — R47 140k — C54 1.8n — U7D TL074 — C53 1n

R55 1k — C76 100n — C79 1n — R50 200k — TP20

R48 1k

+12_VDC_A — C7 100n, C73 100n

U7E TL074

−12_VDC_A — C74 100n, C75 100n

U8 TL071 — R49 1k

R54 1k — C77 100n — C78 1n — TP19 — R51 200k

XLR3 J8

Single ended to differential. Same RC and CR combination as input.

Conn_01x02_Male J7

Output jack J6 — +5_VDC_D

ADC/DAC

U6 AD7680BRJZ−REEL7

+5_VDC_REF — C18 100n — MIC_IN — VDD, VIN, GND — CS*, SDATA, SCLK

U11 AD7680BRJZ−REEL7

+5_VDC_REF — C50 100n — MIC_ER — VDD, VIN, GND — CS*, SDATA, SCLK

U12 AD5541LRZ

+5_VDC_REF — C24 10u, C56 100n — REF, VDD — CS*, DIN, SCLK — VOUT — DAC — DGND, AGND

ADCs: VDD = Vref (need to be very stable). Range from GND up to Vref. 16 bit SAR.

DAC: 16 bit voltage output.

Daniel Kopiecki

Sheet: /

File: in_input.sch

Title: Active noise reduction development board

Size: A3	Date: 2022−11−20	Rev: 0.1
KiCad E.D.A. kicad 5.1.9+dfsg1−1+deb11u1		Id: 1/1